

MODEL QUESTION PAPER 1

S6 INSTRUMENTATION ENGINEERING

COURSE: MECHATRONICS AND CONTROL

Time: 3 Hour

Max. Marks: 75

I: Part –A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	What is the SI unit of flow rate of a liquid or fluid? a) m ³ /s b)m ² /s c)m/s ² d) m/s	M1.03	R
2	Define the term mechatronics.	M1.01	R
3	A relay is switch. a) a mechanical b) an electronic c) an electromechanical d) none of these.	M2.02	R
4	An OR function implemented in ladder logic uses: a) Normally-closed contacts in series b) Normally-open contacts in series c) Normally-open contacts in parallel d) Normally-closed contacts in parallel.	M3.02	U
5	Identify the input device in PLC. a) LED b) Motor c) Lamp d) Sensor	M3.01	R
6	Write the expansion of PLC.	M3.01	R
7	The main function of actuator is a) to produce motion b) detect input c) detect output d) detect the state of the system.	M2.01	U
8	State any two advanced applications in Mechatronics	M4.02	U
9	What is a PLC register?	M3.03	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Draw the working of any flow sensor.	M1.03	U
2.	Describe the importance of mechatronics.	M1.01	U
3.	Describe the temperature monitoring method.	M4.02	U
4.	Describe about Gear trains.	M2.01	U
5.	Draw the PLC ladder diagram for an XOR gate.	M3.02	A
6.	Describe selection of PLC.	M3.01	U
7.	Compare traditional and mechatronics design.	M4.01	U
8.	Describe vibration monitoring method.	M4.02	U
9.	Describe the working of a proximity sensor.	M1.03	U
10.	Draw the block diagram of a Closed loop control system.	M1.02	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain automatic water level controller with a diagram.	M1.02	U
	OR		
IV	Describe the working of pressure, flow and force sensors.	M1.03	U
V	With neat diagram, explain the working of a DC motor	M2.02	U
	OR		
VI	Describe the working of a rotary actuator.	M2.03	U
VII	Explain about solenoids.	M2.02	U
	OR		
VIII	Explain the working of a stepper motor.	M2.02	U
IX	Explain the block diagram of PLC.	M3.01	U
	OR		
X	Draw the PLC ladder diagrams for NAND, NOR and NOT logic functions.	M3.02	U
XI	Explain about latching and sequencing	M3.03	U
	OR		
XII	Explain about timers and counters.	M3.03	U
XIII	Explain about a pick and place robot.	M4.01	U
	OR		
XIV	Explain about car engine management system.	M4.01	U

Question Wise Analysis

Course: **MECHATRONICS**– Question -1

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.03	R	1	2
I.2	M1.01	R	1	2
I.3	M2.02	R	1	2
I.4	M3.02	U	1	2
I.5	M3.01	R	1	2
I.6	M3.01	R	1	2
I.7	M2.01	U	1	2
I.8	M4.02	U	1	2
I.9	M3.03	R	1	2
II.1	M1.03	U	3	8
II.2	M1.01	U	3	8
II.3	M4.02	U	3	8
II.4	M2.01	U	3	8
II.5	M3.02	A	3	8
II.6	M3.01	U	3	8
II.7	M4.01	U	3	8
II.8	M4.02	U	3	8
II.9	M1.03	U	3	8
II.10	M1.02	U	3	8
III	M1.02	U	7	16
IV	M1.03	U	7	16
V	M2.02	U	7	16
VI	M2.03	U	7	16
VII	M2.02	U	7	16
VIII	M2.02	U	7	16
IX	M3.01	U	7	16
X	M3.02	A	7	16
XI	M3.03	U	7	16
XII	M3.03	U	7	16
XIII	M4.01	U	7	16
XIV	M4.01	U	7	16

Course: **MECHATRONICS –QP 1**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	13	26 ± 6	2	2	4	12	2	14	8	28
2	15	31 ± 6	2	2	1	3	4	28	7	33
3	18	37 ± 6	4	4	2	6	4	28	10	38
4	14	29 ± 6	1	1	3	9	2	14	6	24
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	6	4.8
Understanding	107	87.1
Applying	10	8.1
Analysing	0	0
Evaluating	0	0
Creating	0	0
Total	123	100

MODEL QUESTION PAPER 2

S6 INSTRUMENTATION ENGINEERING

SUBJECT: MECHATRONICS AND CONTROL

Time: 3 Hour

Max. Marks: 75

I: Part – A: Answer all the following questions in one word or one sentence. (1 x 9 = 9Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1	Define the term sensor.	M1.03	R
2	What is the principle of operation of proximity sensor?	M1.03	R
3	Write any two types of control systems.	M1.02	R
4	Define the term actuator	M2.01	U
5	What are the types of power supplies?	M2.02	R
6	A normally-open type contact will when its bit is 0	M3.02	R
7	Each horizontal line in ladder diagram is known as	M3.02	R
8	State any two examples of monitoring methods in mechatronicsWhat does PLC stands for?	M4.02	U
9	What does PLC stands for?	M3.01	R

II: Part – B: Answer any eight questions. Each question carries 3 Marks. (3 x 8 = 24 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
1.	Describe the velocity sensor	M1.03	U
2.	What is sequential controller?	M1.02	U
3.	Explain freedom and constraints.	M2.01	U
4.	What are the mechanical aspects in motor selection?	M2.02	U
5.	Draw the PLC ladder diagram for an AND gate.	M3.02	A
6.	Describe about master and jump controls in PLC.	M3.03	U
7.	Describe the working of car park barrier.	M4.01	U
8.	Describe temperature monitoring method.	M4.02	U
9.	Draw a neat sketch of stepper motor.	M2.02	U
10.	Describe a force sensor.	M1.03	U

Part – C

Answer any six questions. Each question carries 7 marks. (7 x 6 = 42 Marks)

		<i>Module Outcomes</i>	<i>Cognitive Level</i>
III	Explain about different types of Control system. OR	M1.02 M1.02	U
IV	With a neat sketch explain automatic water level controller		U
V	Explain different type of mechanical actuation systems OR	M2.01 M2.02	U
VI	Explain different type of electrical actuation systems..		U
VII	Explain the block diagram of PLC. OR	M3.01 M3.04	U
VIII	Explain the selection criteria of PLC.		U
IX	Explain the advanced applications in Mechatronics. OR	M4.01	U
X	Explain vibration and temperature monitoring methods	M4.02	U
XI	Draw the ladder diagrams of NAND, OR and XOR logic functions OR	M3.02	U
XII	Draw the ladder diagrams of NOT, NOR and XNOR logic functions	M3.02	U
XIII	Differentiate Traditional Vs Mechatronics designs. OR	M4.01	U
XIV	Explain about Mechatronics control in automated manufacturing.	M4.01	U

Question Wise Analysis

Course: **MECHATRONICS**– Question -2

Q. No	Module Outcome	Cognitive Level	Marks	Time(Min.)
I.1	M1.03	R	1	2
I.2	M1.03	R	1	2
I.3	M1.02	R	1	2
I.4	M2.01	U	1	2
I.5	M2.02	R	1	2
I.6	M3.02	R	1	2
I.7	M3.02	R	1	2
I.8	M4.02	U	1	2
I.9	M3.01	R	1	2
II.1	M1.03	U	3	8
II.2	M1.02	U	3	8
II.3	M2.01	U	3	8
II.4	M2.02	U	3	8
II.5	M3.02	A	3	8
II.6	M3.03	U	3	8
II.7	M4.01	U	3	8
II.8	M4.02	U	3	8
II.9	M2.02	U	3	8
II.10	M1.03	U	3	8
III	M1.02	U	7	16
IV	M1.02	U	7	16
V	M2.01	U	7	16
VI	M2.02	U	7	16
VII	M3.01	U	7	16
VIII	M3.04	U	7	16
IX	M4.01	U	7	16
X	M4.02	U	7	16
XI	M3.02	A	7	16
XII	M3.02	A	7	16
XIII	M4.01	U	7	16

XIV	M4.01	U	7	16
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Course: **MECHATRONICS AND CONTROL – QP 2**

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Mark Distribution

Module	hr / Module	Marks/Module ($h_i / \sum H_i$) * 123 ($\pm 5\%$)	Type of Questions							
			Part A		Part B		Part C		Total	
			No of Questions	Marks	No of Questions	Marks	No of Questions	Marks	No of Questions	Marks
1	13	26 ± 6	3	3	3	9	2	14	8	26
2	15	31 ± 6	2	2	3	9	2	14	7	23
3	18	37 ± 6	3	3	2	6	4	28	10	34
4	14	29 ± 6	1	1	2	6	4	28	6	34
Total	60	123	9	9	10	30	12	84	31	123

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Cognitive Level Mark Distribution

Cognitive Level	Marks	% of Marks
Remembering	7	6
Understanding	113	92
Applying	3	2
Analysing	0	0
Evaluating	0	0
Creating	0	0

Total	123	100
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